

ERYTHROPOIESIS

- **Erythropoiesis** is the process of origin, development & maturation of erythrocytes.
- **Hemopoiesis / Hematopoiesis** → Process of origin, development & maturation of all blood cells.

SITE OF ERYTHROPOIESIS

- a. **In Fetal Life :—**
 - i. **Mesoblastic Stage:** From mesenchyme of yolk sac (*during 1st two months of intrauterine life*).
 - ii. **Hepatic Stage:** From liver → 3rd month of IUL (*spleen & lymphoid organs also involved*).
 - iii. **Myeloid Stage:** From red bone marrow & liver (*during last 3 months of IUL*).
- b. **In Newborn Babies, Children & Adults :—**
 - *Only from red bone marrow.*
 - i. **Upto age of 20 years** → From red bone marrow of all bones.
 - ii. **After age of 20 years** → From membranous bones (*vertebra, sternum, ribs, scapula & ends of long bones*).

The shaft of long bones → Becomes yellow bone marrow due to fat deposition & loses erythropoietic function.

PROCESS OF ERYTHROPOIESIS

Stem Cells

- These are primary cells capable of **self-renewal and differentiating** into specialized cells.
- **Hemopoietic stem cells (in bone marrow)** → Give rise to blood cells.
- **Uncommitted Pluripotent Hemopoietic Stem Cells (PHSC):** Hemopoietic stem cells in bone marrow which give rise to all types of blood cells.
- **Committed PHSC:** When cells are designed to form a particular type of blood cell.

STEM CELLS LINEAGE & DIFFERENTIATION

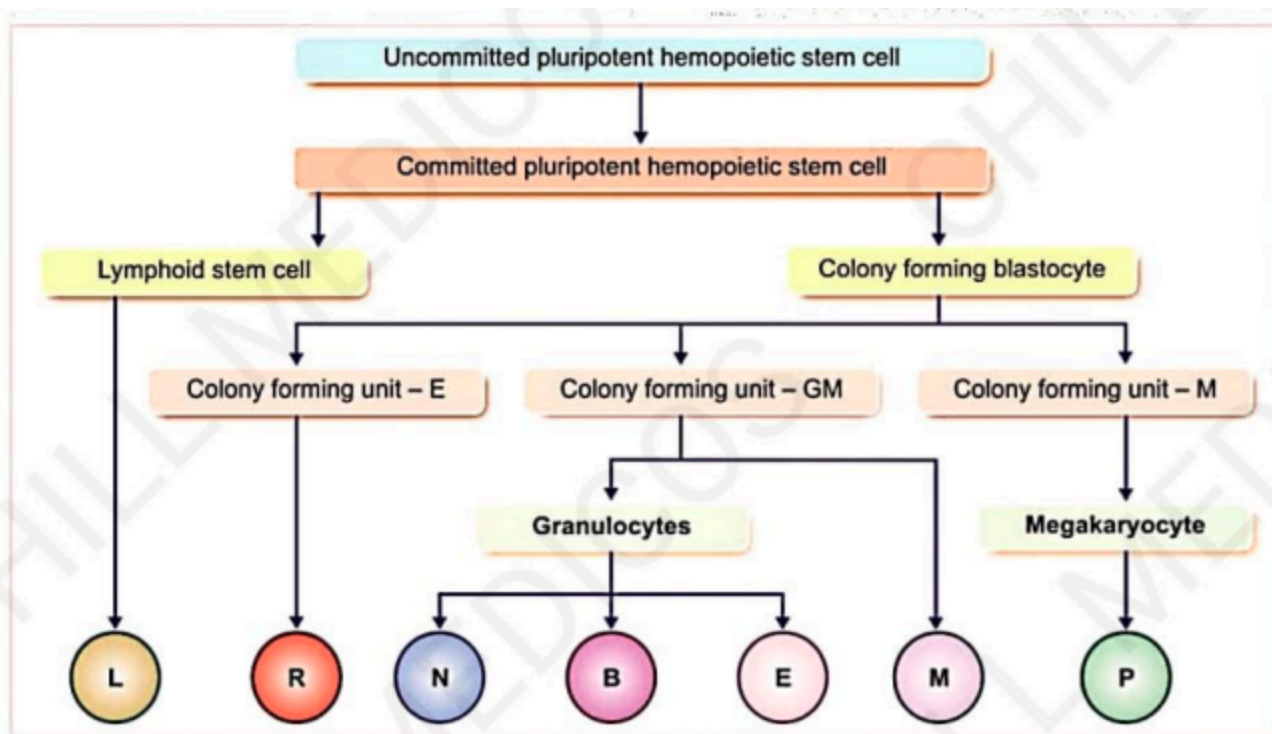
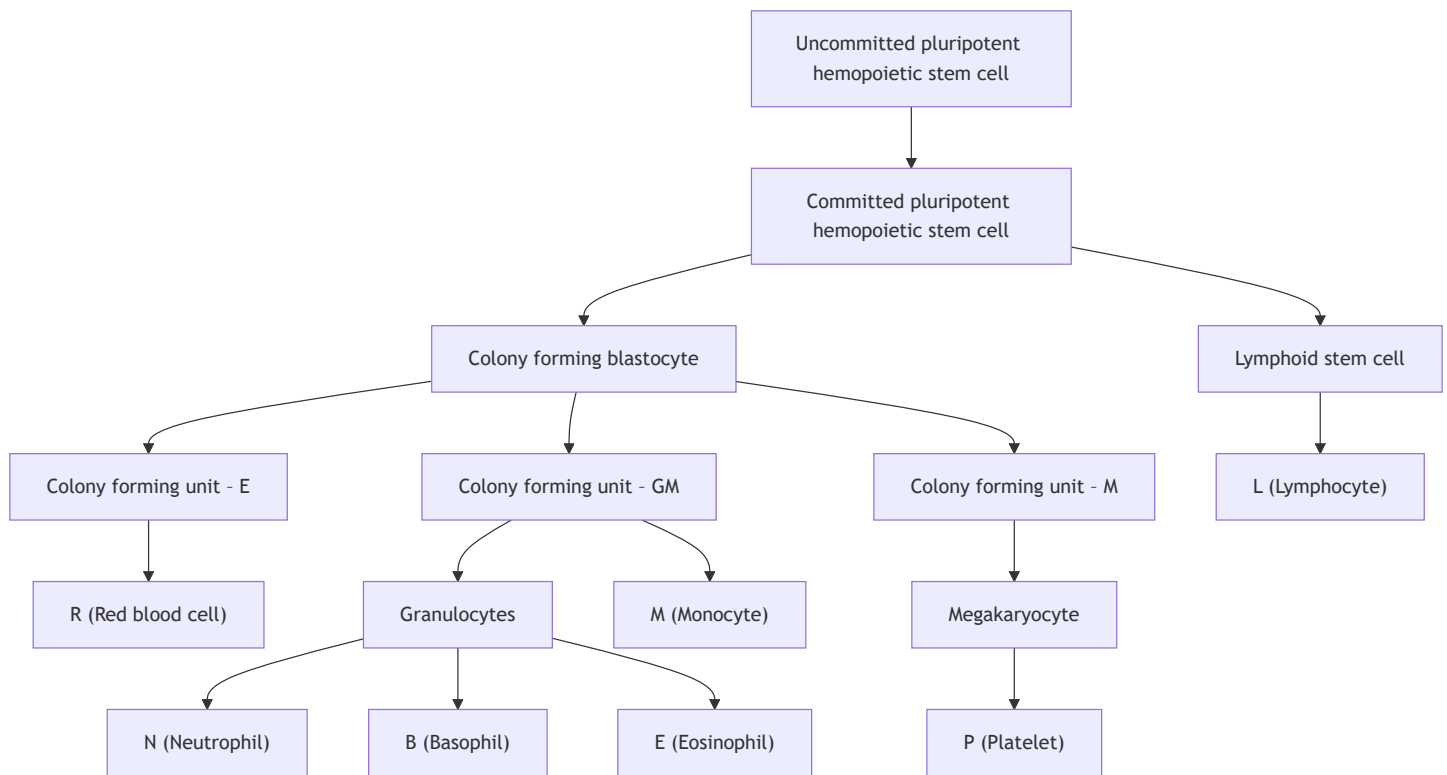


FIGURE 10.1: Stem cells. L = Lymphocyte, R = Red blood cell, N = Neutrophil, B = Basophil, E = Eosinophil, M = Monocyte, P = Platelet.

CHANGES DURING ERYTHROCYTE FORMATION

- a. Reduction in size of the cell.
- b. Disappearance of nucleoli & nucleus.
- c. Appearance of hemoglobin (Hb).
- d. Change in staining properties of cytoplasm.

STAGES OF ERYTHROPOIESIS

Various stages between CFU-E cells and matured RBCs are :—

CFU-E

7 days

1. Proerythroblast
(Proerythrocyte /
Megaloblast)

2. Early Normoblast

3. Intermediate
Normoblast (Polychromatic
erythroblast)

4. Late Normoblast
(Orthochromic
erythroblast)

5. Reticulocyte (Immature
RBC)

Takes 2 days

6. Matured Erythrocyte

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graph TD; A[CFU-E] -- 7 days --> B["1. Proerythroblast<br/>(Proerythrocyte / Megaloblast)"]; B --> C[2. Early Normoblast]; C --> D["3. Intermediate Normoblast<br/>(Polychromatic erythroblast)"]; D --> E["4. Late Normoblast<br/>(Orthochromic erythroblast)"]; E --> F[5. Reticulocyte (Immature RBC)]; F -- Takes 2 days --> G[6. Matured Erythrocyte];
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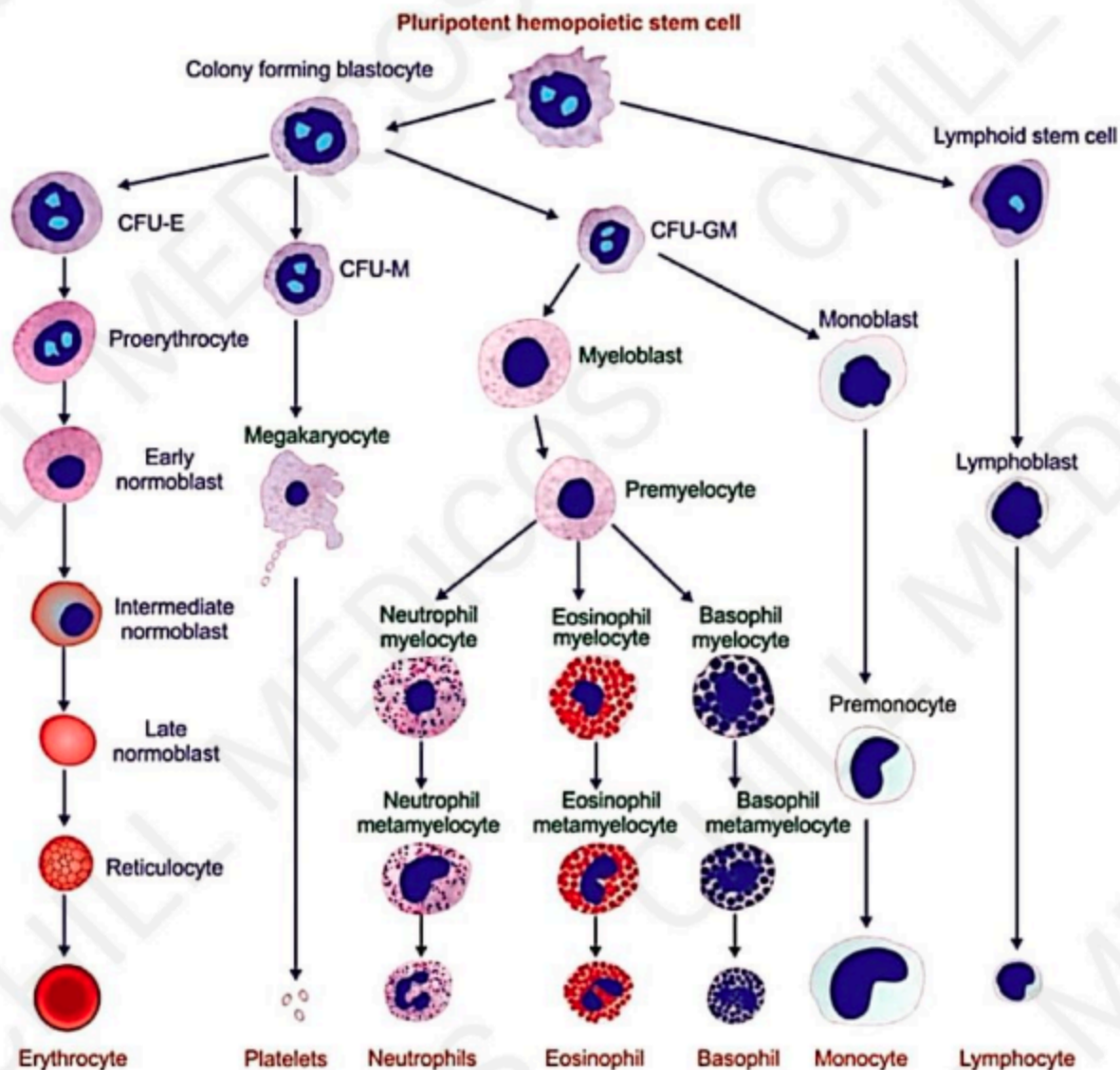


FIGURE 10.2: Stages of erythropoiesis. CFU-E = Colony forming unit-erythrocyte, CFU-M = Colony forming unit-megakaryocyte, CFU-GM = Colony forming unit-granulocyte/monocyte.

IMPORTANT EVENTS DURING ERYTHROPOIESIS

Stage of Erythropoiesis	Important Event
Proerythroblast	Synthesis of hemoglobin starts
Early normoblast	Nucleoli disappear
Intermediate normoblast	Hemoglobin starts appearing
Late normoblast	Nucleus disappears
Reticulocyte	• Reticulum is formed

- Cell enters capillary from site of production |
- | **Matured RBC** | • Reticulum disappears

- Cell attains biconcavity |

DETAILED CHARACTERISTICS OF STAGES

- a. **Proerythrocyte / Proerythroblast (Megaloblast):**
 - Very large in size ($0 - 20 \mu\text{m}$)
 - Nucleus large, occupies cell completely
 - 2 or more nucleoli & reticular network
 - No Hb
 - Cytoplasm is basophilic in nature
- b. **Early Normoblast:**
 - Little smaller ($0 - 15 \mu\text{m}$)
 - Nucleoli disappear
 - Condensation of chromatin (dense)
 - Hb develops
 - Cytoplasm basophilic
- c. **Intermediate Normoblast (Polychromatic Erythroblast):**
 - Smaller ($10 - 12 \mu\text{m}$)
 - Nucleus present
 - Hb appears
 - Stains with both acid + base stains (*Hb takes acid stain, basophilic background takes basic stain*)
- d. **Late Normoblast (Orthochromic Erythroblast):**
 - Size decreases ($8 - 10 \mu\text{m}$)
 - Very much condensed chromatin network (*ink-spot nucleus*)
 - Hb quantity increases
 - Cytoplasm is acidophilic
 - Nucleus disintegrates & disappears
- e. **Reticulocyte (Immature RBC):**
 - Cytoplasm contains reticular network
 - Stains with supravital stain
 - Basophilic (*due to presence of remnants of disintegrated Golgi apparatus & other organelles*)

- Cell transports through capillary membrane (*diapedesis*)
- f. **Matured Erythrocyte:**
 - Reticular network disappears
 - Attains biconcave shape
 - Size decreases ($7.2\ \mu\text{m}$)
 - No nucleus
 - Takes 2 days for maturation